LL

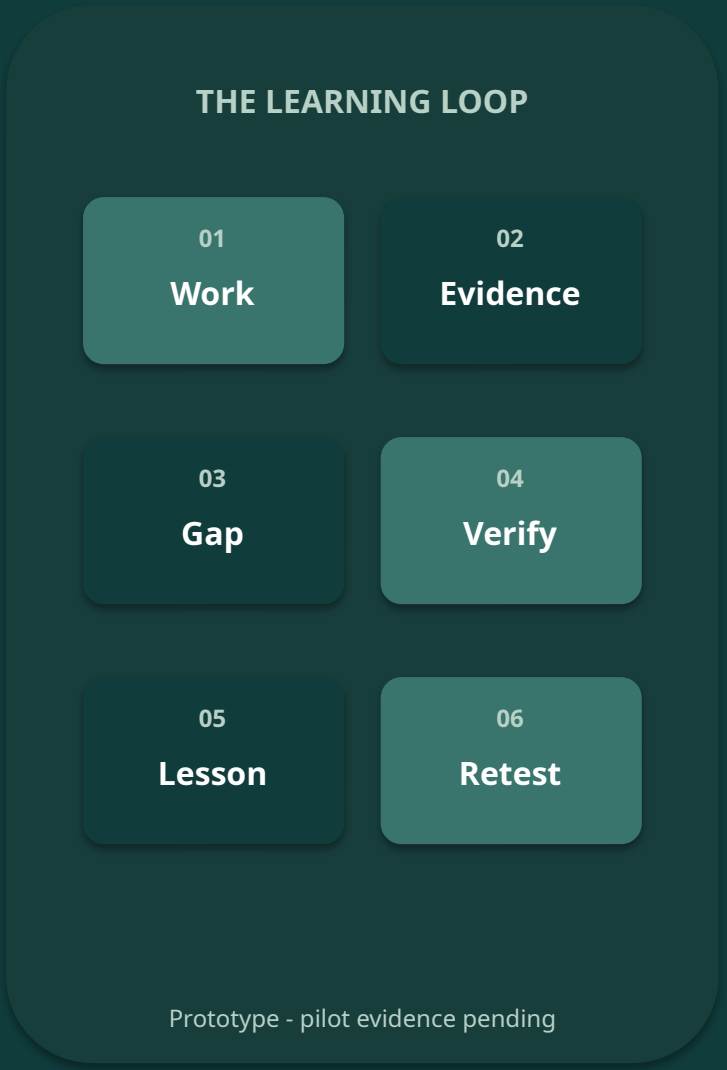
AI FOR EDUCATION

LearnLens AI

From wrong answers
to a clearer next step.

Diagnose · Verify · Teach · Retest

Craft N Code 2026 | Judge presentation



A score tells you what happened. Not always why.

LearnLens targets the moment after a student gets an answer wrong.

FOR STUDENTS

“I lost a mark. What idea did I misunderstand?”

A correction may not show the underlying rule or the next practice step.

FOR TEACHERS

“Which misconception should I reteach first?”

Repeated class patterns need a quick, reviewable way to guide a short intervention.

THE PRODUCT IDEA

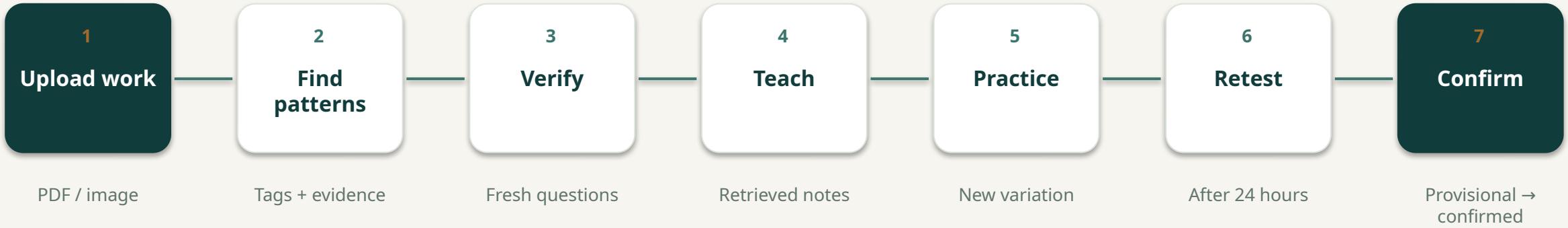
Turn work into a learning loop.

Evidence → a tentative gap → verification → a targeted lesson → a spaced retest.

Designed for students and teachers; not a replacement for teacher judgement or formal grading.

A repeatable path from evidence to follow-up

The diagnosis is a hypothesis. A fresh question checks whether it is useful.



One correct answer is not enough to confirm mastery. A successful retest must be due; early retests stay provisional.

A small architecture designed to degrade clearly

Core retrieval and deterministic fallback can run locally; external AI and persistence are optional.

STUDENT + TEACHER UI

Next.js / TypeScript

- Join by nickname + consent
- Learning loop / read-aloud
- Heatmap / CSV / reteach draft

LEARNING API

FastAPI / Pydantic

- Work extraction + diagnosis
- Misconception taxonomy
- Spaced mastery + deletion

CONTENT + RETRIEVAL

45 notes / 22 tags

- Local BM25-style ranking
- Source IDs + snippets
- No embedding API needed

OPTIONAL SERVICES

Server-side only

- OpenRouter free route
- Gemini optional fallback
- Supabase + SQLite cache

Secrets remain on the server. No live API or database credentials are included in this package.

Features that judges can click through

Implementation status is not the same as evidence of learning impact.

01

Misconception diagnosis

22 taxonomy tags, rule fallback and a verification step.

02

Grounded lessons

45 bundled notes; source IDs/snippets shown; English/Kannada/Hindi options.

03

Spaced mastery

BKT-inspired probability, 14-day half-life and delayed retest gate.

04

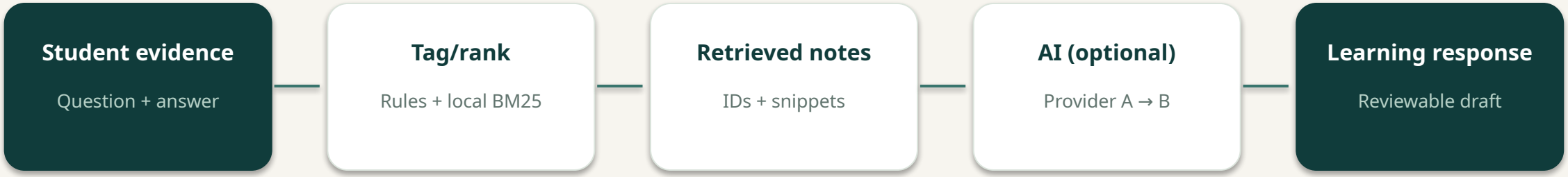
Teacher decisions

Student × concept heatmap, top three tag patterns, 10-minute plan and CSV export.

All diagnosis, translation and reteach suggestions need teacher review; mastery must not be used as an official grade.

Make the system explain where its help comes from

This is real local lexical retrieval over bundled notes, not a claim that a vector database is currently running.



Fallback, not fake certainty

Unsupported offline topics return a clear availability warning.

Fingerprint cache

Results are scoped to the student profile; deletion attempts to purge the profile cache.

Visible provenance

Provider/cache status and retrieved note sources are inspectable in the app/API.

Library: 32 original notes + 13 AI-assisted drafts pending team/educator review; none are independently validated.

We separate a working demo from a validated product

No surveys, pilot results, accuracy benchmark or before/after improvement numbers are invented.

DESIGNED INTO THIS PROTOTYPE

- ✓ Nickname-only class join; no student email/password
- ✓ Consent before class membership is created
- ✓ Privacy page: clear browser profile + request backend/cache deletion
- ✓ Teacher key generated randomly; created key stored hashed
- ✓ Provisional mastery and explicit human-review caveat
- ✓ Synthetic heatmap and worksheet labeled as synthetic

PENDING BEFORE IMPACT CLAIMS

- ○ Student survey + teacher interviews
- ○ 15–20 human-labelled worksheets; inter-rater agreement
- ○ Provider latency, cache hit and failure-rate logging
- ○ Paired pre/post test + delayed retest follow-up
- ○ Hindi/Kannada review with fluent speakers
- ○ Production security and deletion verification

A demo fixture is useful for QA; it is not a real student, a pilot, or a learning outcome.

See the loop. Challenge the assumptions.

01 **Join**

Blue Falcon · CSE-A · consent

02 **Analyse**

Use the clearly synthetic PDF

03 **Learn**

Check note IDs, change language, read aloud

04 **Retest**

Show provisional state; explain 24-hour gate

05 **Teach**

Heatmap, top tags, plan draft and CSV

Next milestone: approved pilot + independently labelled worksheets + production privacy review.

LearnLens AI | Prototype status and all evidence limitations are documented in README.md / docs/evidence.md